

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CL)

November 2012

CL 301 Transportation Engineering-I

Date: 26.11.2012, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Derive an equation for finding the super elevation required if the design coefficient of lateral friction is 'F'. [02]

(b) Draw typical cross section of NH and MDR in rural area. [05]

Q - 2 (a) Write a short note on tar. [03]

(b) Calculate the road length of a district as per Nagpur road plan. [05]

(i) Total area = 6300 km²

(ii) length of railway track = 75 km

(iii) Agricultural area = 2800 km²

No. of villages with population ranges <500, 501-100, 1001-2000, 2001-5000 and above 5001 are 450, 320, 110, 50 and 10 respectively.

Population range	10001 – 20000	20001 – 50000	50001 – 100000	>100000
No. of towns	20	10	6	2

(c) List out various regulatory signs and draw any four. [06]

OR

Q - 2 (a) Write a short note on cutback bitumen. [03]

(b) Give brief description macadam's construction with neat sketch. [05]

(c) List the various warning signs and draw any four. [06]

Q - 3 (a) Enlist the various objectives and uses of traffic volume study. [02]

(b) Which are the measures for reducing the accident? Explain any one. [04]

(c) Do as directed:

(i) A vertical summit curve is to be designed when two grades, +1/50 and -1/80 meet on a highway. The stopping sight distance and overtaking sight distance required are 190 and 600 m respectively. Due to site conditions the length of vertical curve has to be restricted to a maximum value of 500 m. calculate the

length of summit curve needed to fulfill the requirements of (i) Stopping sight distance, (ii) overtaking sight distance. Comment on results.

(ii) Design the geometric features [05]

(a) super elevation

(b) extra widening of pavement

(c) Length of transition curve.

Data available are, design speed = 60 kmph, length of wheel base = 6m, pavement width = 10.5 m, radius of horizontal circular curve = 325 m.

OR

Q - 3 (a) Write the method to measure a spot speed study with neat sketch. [02]

(b) Enlist the methods of O & D studies and explain any one of them. [04]

(c) Do as directed:

(i) Calculate the minimum sight distance required to avoid a head-on collision of two vehicles approaching from opposite directions at 60 and 95 kmph. Reaction time is 2.5 seconds, coefficient of friction = 0.7 and brake efficiency = 50%. [03]

(ii) Explain PIEV theory with neat sketch [05]

SECTION - II

Q - 4 (a) Write a note on construction of Earthen / WBM Roads. [04]

(b) On which factors the selection of Base course and Surface course depends?

List out the stages of Engineering Surveys. [03]

Q - 5 (a) Give the requirements of highway drainage. [04]

(b) Enlist various types of failures in flexible pavement and explain frost heaving [05]

(c) Write a note on various methods of Economic Analysis of Highways. [05]

OR

Q - 5 (a) Write a note on 'Lowering of Water Table' method of sub-surface drainage with neat diagram. [04]

(b) Enlist various types of cracks failures in rigid pavement and explain scalling of concrete. [05]

(c) Write a note on Highway Finance. [05]

Q - 6 (a) Write the basic requirements of an ideal alignment between two terminal stations. [03]

(b) It is proposed to widen an existing 2 lane National Highway section to 4 lane divided road. Design the pavement for new carriageway with the following data: [07]

i) 4 lane divided carriageway

ii) Initial Traffic in each directions in the year of completion of construction

= 5600 CVPD

- iii) Design Life = 10 years
 - iv) Design CBR of Subgrade soil = 5 %
 - v) Traffic Growth Rate = 8 %
 - vi) Vehicle Damage Factor = 4.5.
- (c) Compute the design traffic (N) in terms of million standard axles for two-lane single carriageway with initial traffic of 400 CVPD (sum of both directions), traffic growth rate per annum as 7.5 % and design life as 15 years. [04]

OR

Q - 6 (a) Write a note on Obligatory Points governing the highway alignment. [03]

- (b) Determine the warping stress at interior, edge and corner regions in a 25 cm thick concrete pavement with transverse joints at 11 m interval and longitudinal joints at 3.6 m intervals. The modulus of subgrade reaction (K) is 6.9 kg/cm^3 . Assume temperature differential for day conditions to be 0.6°C per cm slab thickness. Assume radius of loaded area as 15 cm for computing warping stress at the corner. Additional data are given below: [07]

$$e = 10 \times 10^{-6} \text{ per } ^\circ\text{C}$$

$$E = 3 \times 10^5 \text{ kg/cm}^2$$

$$\mu = 0.15$$

- (c) Compute the radius of relative stiffness of 15 cm thick CC slab from the following data: [04]

- i) Modulus of Elasticity of CC = 210000 kg/cm^2
- ii) Poisson's Ratio for concrete = 0.15
- iii) Modulus of Subgrade reaction, K = a) 3.0 kg/cm^3 and b) 7.5 kg/cm^3 .
